

Interview Questions and Answers

Q1: What is the purpose of the `np.linspace` function in NumPy?

A1: The `np.linspace` function in NumPy is used to return evenly spaced numbers over a specified interval. It generates a sequence of numbers starting from `start` to `stop` (inclusive if `endpoint=True`) with a specified number of samples (`num`).

Q2: How does the `np.random.rand` function generate random numbers?

A2: The `np.random.rand` function generates random floats in the half-open interval $[0.0, 1.0)$. It takes dimensions as arguments and returns an array of the specified shape filled with random floats.

Q3: What is the difference between `np.random.randint` and `np.random.rand`?

A3: `np.random.randint` generates random integers from `low` (inclusive) to `high` (exclusive), while `np.random.rand` generates random floats in the half-open interval $[0.0, 1.0)$. The former is used for integer values within a specified range, and the latter is used for generating random floating-point numbers.

Q4: Explain the purpose of the `np.random.seed` function.

A4: The `np.random.seed` function sets the seed for the random number generator in NumPy, ensuring that the sequence of random numbers is reproducible. By setting the seed, you can generate the same sequence of random numbers every time the code is run.

Q5: What does the `shape` attribute of a NumPy array return, and how is it used?

A5: The `shape` attribute of a NumPy array returns a tuple representing the dimensions of the array. It is used to determine the number of elements in each dimension of the array, which is useful for understanding the structure and size of the array.